INVERTIS UNIVERSITY BAREILLY
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BEE-101/BEE-201- Basic Electrical Engineering

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Teaching Scheme: (Lectures - 3 hours/week; Tutorial - 1 hour/week; Laboratory – 2 hours/week)

Course Outcomes

- To understand and analyze basic Electric and Magnetic circuits
- To study the working principles of Electrical Machines and Power Converters.
- To introduce components of Low Voltage Electrical Installations

MODULE-I

D.C. Circuits: Electrical circuit elements (R, L and C), voltage and current sources, Kirchhoff current and voltage laws, analysis of simple circuits with dc excitation. Superposition, Thevenin, Maximum Power Transfer and Norton Theorems. Time-domain analysis of first-order RL and RC circuits.

A.C. Circuits: Representation of sinusoidal waveforms, peak and rms values, phasor representation, real power, reactive power, apparent power, power factor, Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel), resonance. Three-phase balanced circuits, voltage and current relations in star and delta connections.

MODULE-II

Transformers: Magnetic materials, BH characteristics, ideal and practical transformer, equivalent circuit, losses in transformers, regulation and efficiency. Auto-transformer and three-phase transformer connections.

Electrical Machines: Generation of rotating magnetic fields, Construction and working of a three-phase induction motor, Significance of torque-slip characteristic. Loss components and efficiency, starting and speed control of induction motor. Single-phase induction motor.
Construction, working, torque-speed characteristic and speed control of separately excited dc motor. Construction and working of synchronous generators.

MODULE-III

Power Converters: DC-DC bucks and boosts converters, duty ratio control. Single-phase and three-phase voltage source inverters; sinusoidal modulation.

Electrical Installations: Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthing. Types of Batteries, Important Characteristics for Batteries. Elementary calculations for energy consumption, power factor improvement and battery backup.

Text Books:

1. *Basic Electrical Engineering - D.P. Kothari and I.J. Nagrath, 3rd edition 2010, Tata McGraw Hill*
2. *Basic Electrical Engineering - D.C. Kulshreshtha, 2009, Tata McGraw Hill.*

3. *Fundamentals of Electrical Engineering, L.S. Bobrow, Oxford University Press, 2011*
4. *Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010*
5. *Electrical Engineering Fundamentals, Vincent Deltoro, Second Edition, Prentice Hall India, 1989*

Reference Books:

1. *S.N. Singh, "Basic Electrical Engineering" Prentice Hall International*
2. *I.J. Nagarath, " Basic Electrical Engineering" Tata McGraw Hill*
3. *Edward Hughes, " Electrical Technology" Longman*
4. *T.K. Nagsarkar & M.S. Sukhija, "Basic Electrical Engineering" Oxford University Press.*
5. *H. Cotton, " Advanced Electrical Technology" Wheeler Publishing*
6. *W.H. Hayt & J.E. Kennely, "Engineering Circuit Analysis" Mc Graw Hill.*
7. *D.E. Fitzgerald & A. Grabel Higginbotham, " Basic Electrical Engineering Mc- Graw Hill*